

FINAL INSIGHTS DATA MODELLING & TRANSFORMATION USING POWER BI

Project Title: INVENTORY SALES ANALYSIS

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Dataset: Ecommerce retail dataset. In this dataset we have total six table:

- 1) List of Orders
- 2) Order Details
- 3) Sales Target
- 4) Orders Data
- 5) Order Details Duplicate
- 6) Sales Target Duplicate

Key fields:

Order Id, Order date, Customer name, State, City, Location, Category, Sub-Category, Amount, Profit, Quantity, Profit Margin, Profit status, Order Count, Average Profit, Total Amount, Month of Order Date, Target, Month, Total Target.

Data Preparation: Data import, Power Query transformations

- **Data Cleaning: Formatting, null handling, deduplication**

Handling Missing Data & Duplicate Data

CLEAN FUNCTION: Removes line breaks, charterers and other hidden control.

TRIM FUNCTION: This removes non printable characters and extra spaces at the same time.

PROPER FUNCTION: The text can replace capitalized each word.

NULL FUNCTION: you can replace null values using find and replace using replace values options.

For sales data, you can use Power Query to remove blanks, standardize customer names, group by order data using count, average, sum functions, and calculate total Amount before loading the result into Excel. That makes reporting much easier and more reliable.

● Data Integration: Table merging and relationships

Column Creation:

- 1) Merge the City and State columns to create a new column named Location in the format: **City, State**

Screenshot of Location Table:

Answer:

The screenshot shows the Power Query Editor interface. The formula bar contains the following M code: `Table.AddColumn(#"Changed Type1", "Location", each Text.Combine([City], [State]), ", ", type text)`. The data table below shows the result of this operation, with a new 'Location' column added to the existing columns: Order ID, Order Date, Customer Name, State, and City. The 'Location' column contains the concatenation of the 'City' and 'State' columns, separated by a comma and a space.

	Order ID	Order Date	Customer Name	State	City	Location
1	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad	Ahmedabad,Gujarat
2	B-25602	01-04-2018	Pearl	Maharashtra	Pune	Pune,Maharashtra
3	B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal	Bhopal,Madhy Pradesh
4	B-25604	03-04-2018	Divsha	Rajasthan	Jaipur	Jaipur,Rajasthan
5	B-25605	05-04-2018	Kasheen	West Bengal	Kolkata	Kolkata,West Bengal
6	B-25606	06-04-2018	Hazel	Karnataka	Bangalore	Bangalore,Karnataka
7	B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir	Kashmir,Jammu and Kashmir
8	B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu
9	B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow	Lucknow,Uttar Pradesh
10	B-25610	09-04-2018	Yogesh	Bihar	Patna	Patna,Bihar
11	B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram	Thiruvananthapuram,Kerala
12	B-25612	12-04-2018	Shrichand	Punjab	Chandigarh	Chandigarh,Punjab
13	B-25613	12-04-2018	Mukesh	Haryana	Chandigarh	Chandigarh,Haryana
14	B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla	Simla,Himachal Pradesh
15	B-25615	15-04-2018	Bhavina	Sikkim	Gangtok	Gangtok,Sikkim
16	B-25616	15-04-2018	Kanak	Goa	Goa	Goa,Goa
17	B-25617	17-04-2018	Sagar	Nagaland	Kohima	Kohima,Nagaland
18	B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad	Hyderabad,Andhra Pradesh
19	B-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad	Ahmedabad,Gujarat
20	B-25620	20-04-2018	Sarita	Maharashtra	Pune	Pune,Maharashtra
21	B-25621	20-04-2018	Deepak	Madhya Pradesh	Bhopal	Bhopal,Madhy Pradesh
22	B-25622	22-04-2018	Monisha	Rajasthan	Jaipur	Jaipur,Rajasthan
23	B-25623	22-04-2018	Atharv	West Bengal	Kolkata	Kolkata,West Bengal
24	B-25624	22-04-2018	Vini	Karnataka	Bangalore	Bangalore,Karnataka
25	B-25625	23-04-2018	Pinky	Jammu and Kashmir	Kashmir	Kashmir,Jammu and Kashmir
26	B-25626	23-04-2018	Bhishm	Maharashtra	Mumbai	Mumbai,Maharashtra
27	B-25627	23-04-2018	Hitika	Madhya Pradesh	Indore	Indore,Madhy Pradesh
28	B-25628	24-04-2018	Pooja	Bihar	Patna	Patna,Bihar
29	B-25629	24-04-2018	Hemant	Kerala	Thiruvananthapuram	Thiruvananthapuram,Kerala
30	B-25630	24-04-2018	Sahil	Punjab	Chandigarh	Chandigarh,Punjab

- 2) Create a custom column named **Profit Margin** calculated as: **Profit / Amount** (expressed as a percentage)

Screenshot of Profit Margin Table:

Answer:

Power BI Assignment 1-Data Transformation & Data Modeling

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Export query results Refresh Preview Advanced Editor Choose Columns Remove Columns Keep Rows Remove Rows Split Column Group By Data Type: Percentage Merge Queries Append Queries Combine Files Combine

Queries [6]

List of Orders
Order Details
Sales target
Orders Data
Sales target Duplicate
Order Details Duplicate

fx = Table.TransformColumnTypes(#"Added Custom",{"Profit Margin", Percentage.Type})

	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin
1	1,275.00	-1148	7	Furniture	Bookcases	-90.04%
2	66.00	-12	5	Clothing	Stole	-18.18%
3	8.00	-2	3	Clothing	Hankerchief	-25.00%
4	80.00	-56	4	Electronics	Electronic Games	-70.00%
5	168.00	-111	2	Electronics	Phones	-66.07%
6	424.00	-272	5	Electronics	Phones	-64.15%
7	2,617.00	1151	4	Electronics	Phones	43.98%
8	561.00	212	3	Clothing	Saree	37.79%
9	119.00	-5	8	Clothing	Saree	-4.20%
10	1,355.00	-60	5	Clothing	Trousers	-4.43%
11	24.00	-30	2	Furniture	Chairs	-125.00%
12	193.00	-166	3	Clothing	Saree	-86.01%
13	180.00	9	3	Clothing	Trousers	2.78%
14	116.00	16	4	Clothing	Stole	13.79%
15	107.00	36	6	Clothing	Stole	33.64%
16	12.00	1	2	Clothing	Hankerchief	8.33%
17	38.00	18	2	Clothing	Kurti	47.37%
18	65.00	17	2	Clothing	T-shirt	26.15%
19	157.00	5	9	Clothing	Saree	3.18%
20	75.00	0	7	Clothing	Saree	0.00%
21	87.00	4	2	Clothing	Shirt	4.60%
22	50.00	15	4	Clothing	Leggings	30.00%
23	1,364.00	-1864	5	Furniture	Tables	-136.66%
24	476.00	0	3	Furniture	Chairs	0.00%
25	257.00	23	5	Clothing	Hankerchief	8.95%
26	856.00	385	6	Electronics	Printers	44.98%
27	485.00	29	4	Electronics	Electronic Games	5.98%
28	25.00	-5	4	Clothing	Saree	-20.00%
29	1,076.00	-38	4	Electronics	Printers	-3.53%

Query Settings

PROPERTIES

Name

Order Details

All Properties

APPLIED STEPS

Source

Promoted headers

Changed column type

Filtered Rows1

Cleaned Text

Trimmed Text

Cleaned Text1

Trimmed Text1

Changed Type

Added Custom

Changed Type1

Added Conditional Column

Conditional Column:

- 1) Add a conditional column named **Profit Status** based on:
 - Profit < 0 → *Loss*
 - Profit = 0 → *Break-Even*
 - Profit > 0 → *Profit*

Screenshot of Profit Status Table:

Answer:

Power BI Assignment 1-Data Transformation & Data Modeling

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Enter Data Data source settings Manage Parameters Export query results Refresh Preview Advanced Editor Choose Columns Remove Columns Keep Rows Remove Rows Split Column Group By Data Type: Any Use First Row as Headers Replace Values Merge Queries Append Queries Combine Files Combine

Queries [6]

- List of Orders
- Order Details
- Sales target
- Orders Data
- Sales target Duplicate
- Order Details Duplicate

fx = Table.AddColumn(#"Changed Type1", "Profit Status", each if [Profit] < 0 then "Loss" else if [Profit] = 0 then "Break-Even" else if

	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status
1	1,275.00	-1148	7 Furniture	Bookcases	-90.04%	Loss
2	66.00	-12	5 Clothing	Stole	-18.18%	Loss
3	8.00	-2	3 Clothing	Hankchief	-25.00%	Loss
4	80.00	-56	4 Electronics	Electronic Games	-70.00%	Loss
5	168.00	-111	2 Electronics	Phones	-66.07%	Loss
6	424.00	-272	5 Electronics	Phones	-64.15%	Loss
7	2,617.00	1151	4 Electronics	Phones	43.98%	Profit
8	561.00	212	3 Clothing	Saree	37.79%	Profit
9	119.00	-5	8 Clothing	Saree	-4.20%	Loss
10	1,355.00	-60	5 Clothing	Trousers	-4.41%	Loss
11	24.00	-30	1 Furniture	Chairs	-125.00%	Loss
12	193.00	-166	3 Clothing	Saree	-86.01%	Loss
13	180.00	5	3 Clothing	Trousers	2.78%	Profit
14	116.00	16	4 Clothing	Stole	13.79%	Profit
15	107.00	36	6 Clothing	Stole	33.64%	Profit
16	12.00	1	2 Clothing	Hankchief	8.33%	Profit
17	38.00	18	1 Clothing	Kurti	47.37%	Profit
18	63.00	17	2 Clothing	T-shirt	26.15%	Profit
19	157.00	5	9 Clothing	Saree	3.18%	Profit
20	75.00	0	7 Clothing	Saree	0.00%	Break-Even
21	87.00	4	2 Clothing	Shirt	-4.60%	Profit
22	50.00	15	4 Clothing	Leggings	30.00%	Profit
23	1,364.00	-1864	5 Furniture	Tables	-136.66%	Loss
24	476.00	0	3 Furniture	Chairs	0.00%	Break-Even
25	257.00	23	5 Clothing	Hankchief	8.95%	Profit
26	856.00	385	6 Electronics	Printers	44.98%	Profit
27	485.00	29	4 Electronics	Electronic Games	5.98%	Profit
28	25.00	-5	4 Clothing	Saree	-20.00%	Loss
29	1,076.00	-38	4 Electronics	Printers	-3.51%	Loss
30						

Query Settings

PROPERTIES

Name

Order Details

All Properties

APPLIED STEPS

- Source
- Promoted headers
- Changed column type
- Filtered Rows
- Filtered Rows1
- Cleaned Text
- Trimmed Text
- Cleaned Text1
- Trimmed Text1
- Changed Type
- Added Custom
- Changed Type1
- Added Conditional Column

Merging Data (Joins)

- Merge the *List of Orders* and *Order Details* tables using **Order ID**.
- Name the merged output table as **Orders Data**.

Screenshot of Orders Data Table:

Answer:

Power BI Assignment 1-Data Transformation & Data Modeling

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Enter Data Data source settings Manage Parameters Export query results Refresh Preview Advanced Editor Choose Columns Remove Columns Keep Rows Remove Rows Split Column Group By Data Type: Table Use First Row as Headers Replace Values Merge Queries Append Queries Combine Files Combine

Queries [6]

- List of Orders
- Order Details
- Sales target
- Orders Data
- Sales target Duplicate
- Order Details Duplicate

fx = Table.RenameColumns(Source,{{"Order Details", "Orders Data"}})

	Order Date	CustomerName	State	City	Location	Orders Data
1	01-04-2018	Bharat	Gujarat	Ahmedabad	Ahmedabad,Gujarat	Table
2	01-04-2018	Pearl	Maharashtra	Pune	Pune,Maharashtra	Table
3	03-04-2018	Jahan	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh	Table
4	03-04-2018	Divsha	Rajasthan	Jaipur	Jaipur,Rajasthan	Table
5	05-04-2018	Kasheen	West Bengal	Kolkata	Kolkata,West Bengal	Table
6	06-04-2018	Hazel	Karnataka	Bangalore	Bangalore,Karnataka	Table
7	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir	Kashmir,Jammu and Kashmir	Table
8	08-04-2018	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	Table
9	09-04-2018	Jitesh	Uttar Pradesh	Lucknow	Lucknow,Uttar Pradesh	Table
10	09-04-2018	Yogesh	Bihar	Patna	Patna,Bihar	Table
11	11-04-2018	Anita	Kerala	Thiruvananthapuram	Thiruvananthapuram,Kerala	Table
12	12-04-2018	Shrichand	Punjab	Chandigarh	Chandigarh,Punjab	Table
13	12-04-2018	Mukesh	Haryana	Chandigarh	Chandigarh,Haryana	Table
14	13-04-2018	Vandana	Himachal Pradesh	Simla	Simla,Himachal Pradesh	Table
15	15-04-2018	Bhavna	Sikkim	Gangtok	Gangtok,Sikkim	Table
16	15-04-2018	Kanak	Goa	Goa	Goa,Goa	Table
17	17-04-2018	Sagar	Nagaland	Kohima	Kohima,Nagaland	Table
18	18-04-2018	Manju	Andhra Pradesh	Hyderabad	Hyderabad,Andhra Pradesh	Table
19	18-04-2018	Ramesh	Gujarat	Ahmedabad	Ahmedabad,Gujarat	Table
20	20-04-2018	Sarita	Maharashtra	Pune	Pune,Maharashtra	Table
21	20-04-2018	Deepak	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh	Table
22	22-04-2018	Monisha	Rajasthan	Jaipur	Jaipur,Rajasthan	Table
23	22-04-2018	Atharv	West Bengal	Kolkata	Kolkata,West Bengal	Table
24	22-04-2018	Vini	Karnataka	Bangalore	Bangalore,Karnataka	Table
25	23-04-2018	Pinky	Jammu and Kashmir	Kashmir	Kashmir,Jammu and Kashmir	Table
26	23-04-2018	Bhishm	Maharashtra	Mumbai	Mumbai,Maharashtra	Table
27	23-04-2018	Hitika	Madhya Pradesh	Indore	Indore,Madhya Pradesh	Table
28	24-04-2018	Pooja	Bihar	Patna	Patna,Bihar	Table
29	24-04-2018	Hemant	Kerala	Thiruvananthapuram	Thiruvananthapuram,Kerala	Table
30						

Query Settings

PROPERTIES

Name

Orders Data

All Properties

APPLIED STEPS

- Source
- Renamed Columns
- Sorted Rows
- Filtered Rows

8. Sorting and Filtering Data

1. Sort records in the **Orders Data** table by **Order Date** in **descending order**.

Screenshot of Table:

Answer:

Power BI Assignment 1-Data Transformation & Data Modeling

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Enter Data Data source settings Manage Parameters Export query results Refresh Preview Advanced Editor Choose Columns Remove Columns Keep Rows Remove Rows Split Columns Group By Data Type: Date Use First Row as Headers Replace Values Merge Queries Append Queries Combine Files

Queries [6] List of Orders Order Details Sales target Orders Data Sales target Duplicate Order Details Duplicate

Table.Sort(#"Filtered Rows",{"Order Date", Order.Descending})

	Order Date	CustomerName	State	City	Location	Orders Data
1	31-03-2019	Hitika	Madhya Pradesh	Indore	Indore,Madhya Pradesh	Table
2	30-03-2019	Bhishm	Maharashtra	Mumbai	Mumbai,Maharashtra	Table
3	29-03-2019	Pinky	Jammu and Kashmir	Kashmir	Kashmir,Jammu and Kashmir	Table
4	28-03-2019	Vini	Karnataka	Bangalore	Bangalore,Karnataka	Table
5	28-03-2019	Atharv	West Bengal	Kolkata	Kolkata,West Bengal	Table
6	28-03-2019	Monisha	Rajasthan	Jaipur	Jaipur,Rajasthan	Table
7	27-03-2019	Deepak	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh	Table
8	27-03-2019	Manju	Andhra Pradesh	Hyderabad	Hyderabad,Andhra Pradesh	Table
9	27-03-2019	Ramesh	Gujarat	Ahmedabad	Ahmedabad,Gujarat	Table
10	27-03-2019	Sarita	Maharashtra	Pune	Pune,Maharashtra	Table
11	27-03-2019	Sagar	Nagaland	Kohima	Kohima,Nagaland	Table
12	26-03-2019	Bhavna	Sikkim	Gangtok	Gangtok,Sikkim	Table
13	26-03-2019	Mukesh	Haryana	Chandigarh	Chandigarh,Haryana	Table
14	26-03-2019	Vandana	Himachal Pradesh	Simla	Simla,Himachal Pradesh	Table
15	26-03-2019	Shrichand	Punjab	Chandigarh	Chandigarh,Punjab	Table
16	26-03-2019	Kanak	Goa	Goa	Goa,Goa	Table
17	25-03-2019	Anita	Kerala	Thiruvananthapuram	Thiruvananthapuram,Kerala	Table
18	24-03-2019	Yogesh	Bihar	Patna	Patna,Bihar	Table
19	23-03-2019	Jitesh	Uttar Pradesh	Lucknow	Lucknow,Uttar Pradesh	Table
20	22-03-2019	Kasheen	West Bengal	Kolkata	Kolkata,West Bengal	Table
21	22-03-2019	Divsha	Rajasthan	Jaipur	Jaipur,Rajasthan	Table
22	22-03-2019	Sonakshi	Jammu and Kashmir	Kashmir	Kashmir,Jammu and Kashmir	Table
23	22-03-2019	Hazel	Karnataka	Bangalore	Bangalore,Karnataka	Table
24	22-03-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	Table
25	21-03-2019	Bharat	Gujarat	Ahmedabad	Ahmedabad,Gujarat	Table
26	21-03-2019	Pournamasi	Madhya Pradesh	Indore	Indore,Madhya Pradesh	Table
27	21-03-2019	Pearl	Maharashtra	Pune	Pune,Maharashtra	Table
28	21-03-2019	Jahan	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh	Table
29	20-03-2019	Sweta	Maharashtra	Mumbai	Mumbai,Maharashtra	Table

Query Settings

PROPERTIES

Name

Orders Data

Applied Steps

Source

Renamed Columns

Sorted Rows

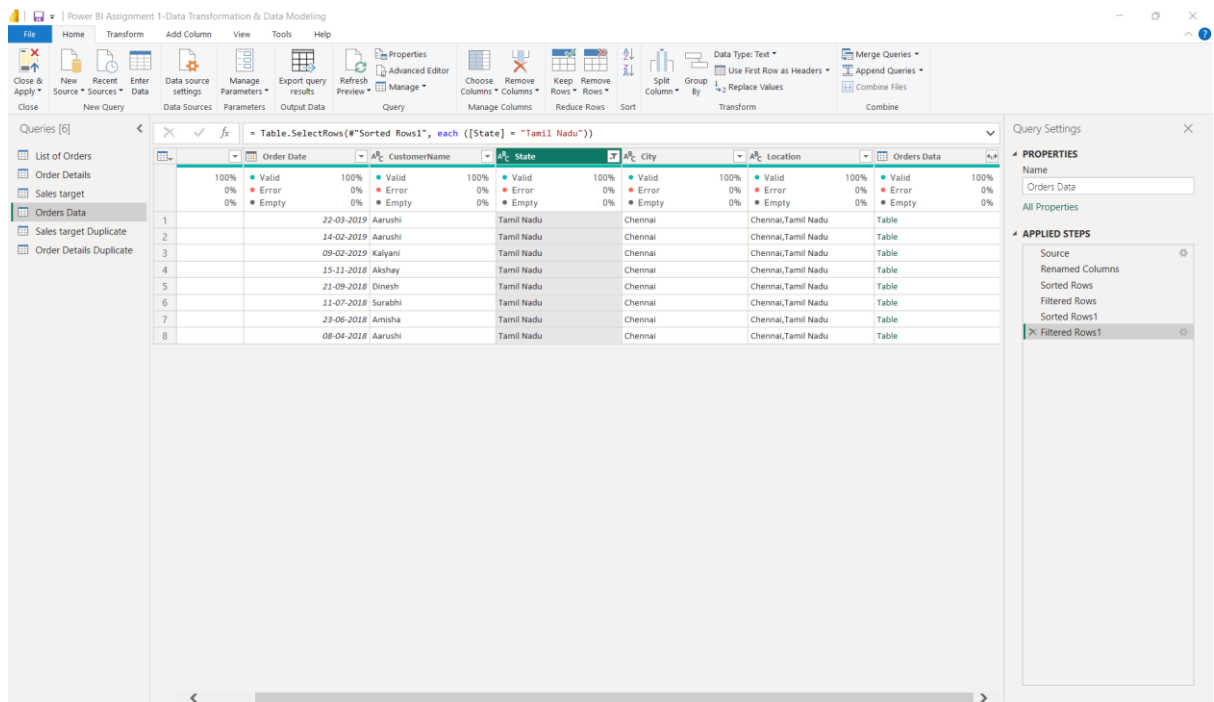
Filtered Rows

Sorted Rows1

2. Apply filters to analyse orders from a specific state (e.g., **Tamil Nadu**).

Screenshot of Table:

Answer:



● Data Aggregation: Grouping and summarization :

Data aggregation is the process of combining detailed records into summaries based on

- Count of Order ID
- Average Profit by Category
- Total Amount by Sub-Category

Screenshot of Table:

Answer:

Power BI Assignment 1-Data Transformation & Data Modeling

File Home Transform Add Column View Tools Help

Close & Apply * New Source * Recent Sources * Enter Data * Data source settings * Manage Parameters * Export query results * Refresh Preview * Advanced Editor * Choose Columns * Remove Columns * Keep Rows * Remove Rows * Split Column * Group By * Data Type: Text * Use First Row as Headers * Merge Queries * Append Queries * Combine Files * Combine

Queries [6]

- List of Orders
- Order Details
- Sales target
- Orders Data
- Sales target Duplicate
- Order Details Duplicate

Table.Group(*Added Conditional Column", {"Order ID", "Category", "Sub-Category"}, {"Order Count", each Table.RowCount(_), Int64.Type),

Order ID	Category	Sub-Category	Order Count	Average Profit	Total Amount
1	Furniture	Bookcases	2	-1148	2275
2	Clothing	Stole	2	-12	66
3	Clothing	Hankerchief	2	-2	8
4	Electronics	Electronic Games	2	-56	80
5	Electronics	Phones	3	256	3209
6	Clothing	Saree	2	103.5	680
7	Clothing	Trousers	2	-27.5	1535
8	Furniture	Chairs	2	-30	24
9	Clothing	Saree	2	-166	193
10	Clothing	Stole	2	26	223
11	Clothing	Hankerchief	2	2	12
12	Clothing	Kurti	2	18	38
13	Clothing	T-shirt	2	17	65
14	Clothing	Saree	2	5	157
15	Clothing	Saree	2	0	75
16	Clothing	Shirt	2	4	87
17	Clothing	Leggings	2	15	50
18	Furniture	Tables	2	-1864	1364
19	Furniture	Chairs	2	0	476
20	Clothing	Hankerchief	2	23	257
21	Electronics	Printers	2	385	856
22	Electronics	Electronic Games	2	29	485
23	Clothing	Saree	2	-5	25
24	Electronics	Printers	2	-316	1857
25	Clothing	Stole	2	-54	107
26	Electronics	Accessories	2	-55	68
27	Clothing	Saree	2	0	43
28	Furniture	Furnishings	2	-5	30
29	Clothing	Saree	2	-59	160
30	Furniture	Chairs	2	-55	259

Query Settings

PROPERTIES

Name

Order Details Duplicate

APPLIED STEPS

- Source
- Promoted headers
- Changed column type
- Filtered Rows
- Filtered Rows1
- Cleaned Text
- Trimmed Text
- Cleaned Text1
- Trimmed Text1
- Changed Type
- Added Custom
- Changed Type1
- Added Conditional Column
- Grouped Rows

- Duplicate the *Sales Target* table and aggregate the total Target by **Month of order date**.

Screenshot of Table:

Answer:

Power BI Assignment 1-Data Transformation & Data Modeling

File Home Transform Add Column View Tools Help

Close & Apply * New Source * Recent Sources * Enter Data * Data source settings * Manage Parameters * Export query results * Refresh Preview * Advanced Editor * Choose Columns * Remove Columns * Keep Rows * Remove Rows * Split Column * Group By * Data Type: Text * Use First Row as Headers * Merge Queries * Append Queries * Combine Files * Combine

Queries [6]

- List of Orders
- Order Details
- Sales target
- Orders Data
- Sales target Duplicate
- Order Details Duplicate

Table.Group(*"Changed Type", {"Month"}, {"Total Target", each List.Sum([Target]), type nullable number))

Month	Total Target
1 April	31400
2 May	31500
3 June	31600
4 July	33800
5 August	33900
6 September	34000
7 October	36200
8 November	36300
9 December	36400
10 January	42500
11 February	43600
12 March	43800

Query Settings

PROPERTIES

Name

Sales target Duplicate

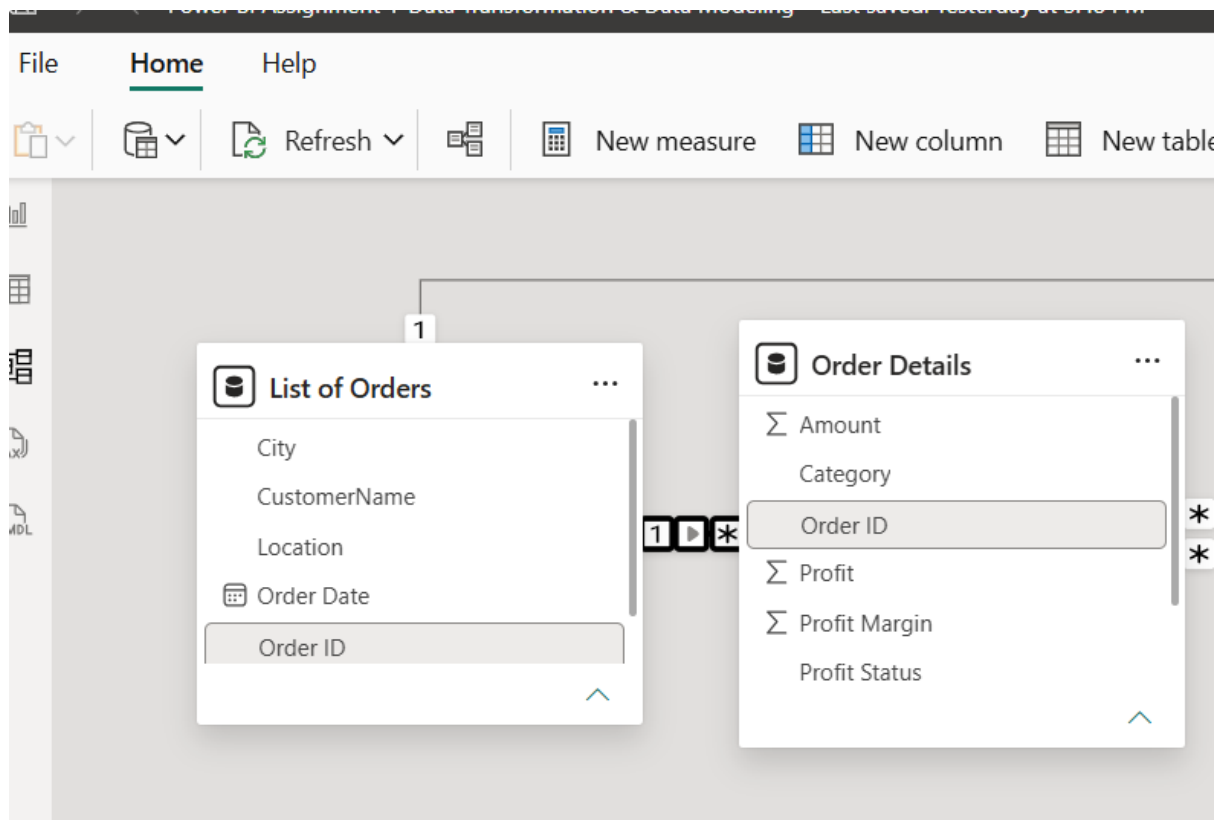
APPLIED STEPS

- Source
- Promoted headers
- Changed column type
- Filtered Rows
- Cleaned Text
- Cleaned Text1
- Duplicated Column
- Renamed Columns
- Extracted Month Name
- Changed Type
- Grouped Rows

● **Data Modeling: Schema design and relationship management.**

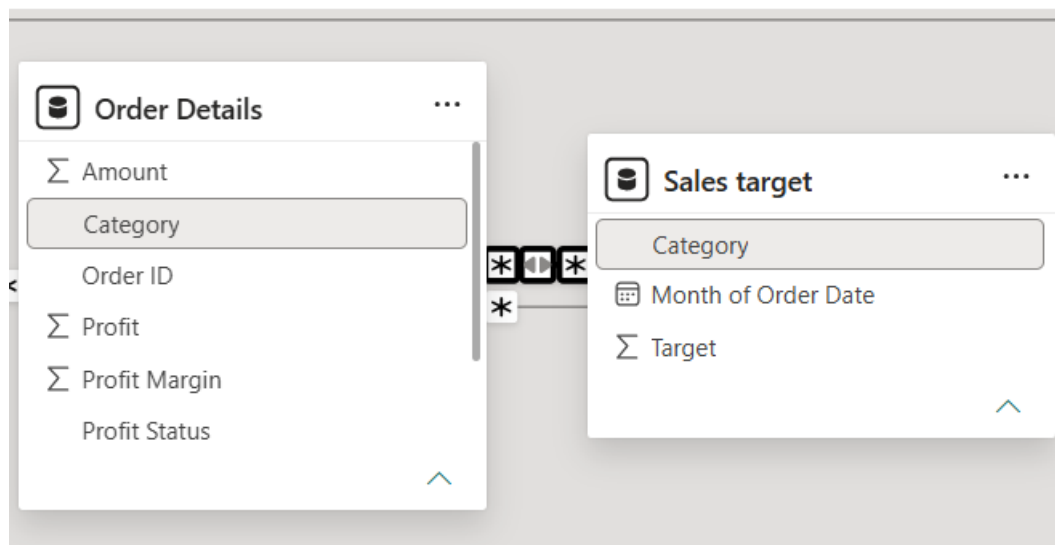
1) Establish a relationship between:

- *List of Orders* and *Order Details* using **Order ID**



1) Establish a relationship between:

- *Order Details* and *Sales Target* using **Category**



- Use **Manage Relationships** to ensure relationships are active and correctly defined.

